

Qno.4 Explanation

```
void copystring(char *dst, const char *src, int n) {
    while(n-- > 0 && *src)
        *dst++ = *src++;
    *dst = '\0';
}
```

Since xv-6 doesn't have a copy function, we make one ourselves. This simple function just take the pointer to the string in src, the number of characters to copy in n, and the destination buffer address in dst.

```
void addCommand(const char *cmd) {
    copystring(history[history_count % HISTORY_SIZE], cmd, CMD_MAX);
    history_count++;
}
```

Simply adds the new command (cmd) to the circular array built to store the command history.

```
// Get previous command
char* get_history() {
    if (history_count == 0)
        return 0;
    history_index--;
    if (history_index < 0)
        history_index = 0;
    return history[history_index % HISTORY_SIZE];
}

// Reset index when new command is typed
void reset() {
    history_index = history_count;
}
```

Function to return the history of commands. If no commands, return 0, else return the command on the current index number. Reset is self explanatory

```

} else if (c == '@') { // go back
    char *prev = get_history();
    if (prev) {
        // Clear current line
        while (i > 0) {
            write(1, "\b \b", 3);
            i--;
        }
        // Copy and show previous command
        copystring(buf, prev, nbuf - 1);
        i = strlen(buf);
        write(1, buf, i);
    }
}

```

If an '@' is detected in the terminal, it pops out the command in the current index.

Current line is cleared to ensure nothing is there on the terminal. Then copystring() is used to put the previous commands, i.e the output of get_history() to the buf (terminal).

In getcmd(), every time something is detected, addcommand() is called to add the command to the history.